

Real-time Person Tracking and Re-identification Based on Feature Matching

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Abstract—The existing video-oriented person tracking and re-recognition research are all based on face recognition, which cannot meet the requirements of high continuity and high accuracy. In order to solve this problem, this paper propose a tracking and re-identification scheme based on KM (Kuhn-Munkras) algorithm. When tracking a person, this scheme establishes the connection between the face image observed in real time, the face database, and the human body image observed in real time to determine their identity. During tracking person, the appearance characteristics can be captured all the time, and the recognition of person has high real-time performance and high continuity. In the tracking process, a real-time updated person information cache list is established to realize the re-identification function after the person tracking is interrupted for a long time.

Keywords—person information cache; KM algorithm; face recognition; face detection; person re-identification; person detection

I. INTRODUCTION

With the rapid development of Internet technology and the continuous advancement of the construction of "Safe City", video surveillance has been widely used and has become the most important security monitoring method at present. In the development process of more than 30 years, the video surveillance system has experienced three stages: analog video surveillance system, digital video surveillance system, and intelligent video surveillance system. Every technological innovation has injected new vitality into the development of video surveillance [1]. At present, we have established computer vision cooperation with the two major video surveillance equipment companies, Haikang and Dahua, which rank first and second in the world by market share.

The face recognition [2] system in video surveillance has a big disadvantage: It often fails to capture facial images that can be used for recognition. On the one hand, under natural circumstances, there are so many person postures that it is impossible to capture the front face all the time. On the other hand, the clarity of images collected in a surveillance environment is usually low, which will affect the extraction of facial features. At present, people try to use the method of person appearance feature fusion to identify person. Face recognition is affected by factors such as distance and clarity, often cannot capture the available face features, but once captured, the facial features is more reliable; and the appearance characteristics will be affected by the wearing

factors of the person, such as people wearing similar clothes, people change the backpack or even the change of clothes will change the appearance features, resulting in unreliable recognition results. But the requirements for the person's posture and images clarity are not high, and it is easy to capture.

DeepSort proposed by Wojke et al. is a detection-based multi-person tracking technology. It has two obvious shortcomings: 1) It only associates the current frame with the previous frame, which can easily lead to target loss due to occlusion. 2) The person in video surveillance can be tracked, but their identity cannot be identified [3]. The Deep Affinity Network (DAN) proposed by Sun et al. uses an end-to-end approach to model the appearance of the target and the correlation between different frames, so as to use deep learning technology for data association to achieve reliable online tracking, but unable to identify the identity of the person [4]. Kim et al. proposed a video-based face recognition and tracking scheme, which solved the problem of tracking and recognizing faces in videos, but did not solve the problem of re-identification of person. And once the face is lost, the tracking and recognition ability will be lost [5]. The face recognition method proposed by Zhao et al. realizes face detection and tracking, but its effect is poor in real environments where the face is occluded, the face is rotated at a large angle, and the illumination changes drastically. And this research has not achieved real-time tracking and identification of multi-person [6]. Although the person tracking and re-recognition system based on facial recognition proposed by Koide et al. achieves real-time tracking and recognition of person, the accuracy of the system's recognition and the continuity of tracking are poor [7].

Aiming at emergencies such as face loss, face overlap, person overlap, and person tracking interruption for a long time that cannot be solved by the above methods, we proposes a real-time person tracking and re-recognition scheme based on face and appearance feature matching. It can maintain high accuracy of re-identification and high continuity of tracking under the above-mentioned emergencies. The person re-identification scheme proposed in this paper is oriented to practical applications and can better adapt to various situations that are prone to occur in video surveillance. And we develop a multi-person real-time tracking and identification system called *personTrack*.

II. MULTI-PERSON REAL-TIME TRACKING AND RE-IDENTIFICATION SCHEME

This section will introduce the solution from five aspects: face detection and feature extraction, human body detection and feature extraction, multi-person matching, tracking and re-identification. Figure 1 is a flowchart of the tracking and re-identification scheme proposed in this paper.

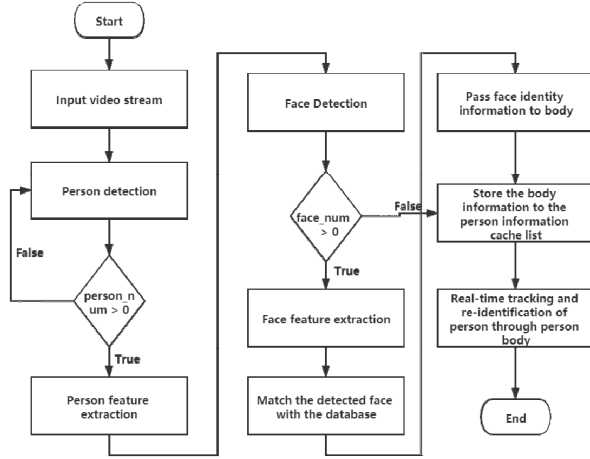


Figure 1. Flow chart of real-time person tracking and re-identification system based on feature matching.

A. Face detection and feature extraction

We use OpenCV to process the video stream, and the video stream is processed into a sequence of images by way of frame extraction. In addition, we use the face data set MS-Celeb-1M to train the face recognition model ArcFace [8], use the data set COCO2017 to train the human body detection model Yolov4 [9], and use the data set WIDER FACE to train the face detection model Retinaface [10].

First, we input the image sequence into the person detector constructed based on Yolov4 to obtain the person image list and the person rectangular frame matrix; input person image into the face detector constructed based on Retinaface to obtain the face image list and the face rectangular frame matrix. Input face image into a face feature extractor based on ArcFace to obtain face feature matrix, each row represents a 512 dimensions face feature of one person.

B. Human body detection and appearance feature extraction

We use pedestrian re-recognition model OSNet [11] to construct the person appearance feature extractor. First, input the current frame of the video into the human body detector built by Yolov4 to obtain the human body images list and the position of the person in the images: p_boxes. Then judge whether p_boxes exists, and if so, pass human body images into the appearance feature extractor to obtain the appearance feature matrix. If p_boxes does not exist, continue processing the next frame.

C. Multi-person matching based on KM algorithm

In the re-identification problem, how to establish a matching relationship between the unknown person with the known person is a difficult problem. The simplest way of multi-person matching is to calculate the distance between all person in pairs set to $D\{d_1, d_2, \dots, d_i, \dots\}$ (i is the serial number of the person), and set a threshold: *Threshold*, when the distance between one pair of targets is less than *Threshold*, the match is considered success. The logic of this method is very simple, but the shortcomings are also very obvious: 1) When faced with a large number of person, there may be multiple same distances and person matching cannot be performed. 2) It is impossible to effectively distinguish multiple similar distances based only on the numerical value.

KM algorithm is used to solve the optimal matching problem in the bipartite graph. KM algorithm can be implemented using the linear assignment function in the machine learning toolkit sklearn. We uses the KM algorithm to match person with known identities and person with unknown identities with high accuracy.

Suppose there are m sets of person with known identities $M\{P_0, P_1, \dots, P_i, \dots, P_{m-1}\}$ and n sets of person with unknown identities $N\{T_0, T_1, \dots, T_j, \dots, T_{n-1}\}$ (i, j are the serial numbers of the person), construct the target set $O\{M, N\}$ to be matched. Calculate the Euclidean distance d_{ij} between each person in M and each person in N , and get the distance matrix

$$D = \begin{bmatrix} d_{00} & \dots & d_{0(n-1)} \\ \vdots & \ddots & \vdots \\ d_{(m-1)0} & \dots & d_{(m-1)(n-1)} \end{bmatrix}. \text{ Now it is necessary to match}$$

each person in M to each person in N , and the sum of the distances is the smallest. This paper takes matrix D as input and passes it into the linear_assignment function, and the return value is the list of best matches. For example, the element (0,3) in L indicates that the person P_0 in M matches the person T_3 in N , and the Euclidean distance between them is d_{03} .

D. Person identification and tracking

In order to identify person in video surveillance, this paper uses the face detection and feature extraction methods in section A to create a person face feature database with name tags. Then use the KM algorithm to establish a matching relationship between the face images in the current frame of the video and the face database to obtain the personal identity information corresponding to the face images.

Because the face rotation amplitudes of the captured face images are different, when the rotation amplitude is too large, the accuracy of face recognition will be greatly reduced. Therefore, we have designed a face rotation amplitude determination algorithm. When the face rotation amplitude is greater than the *Threshold* (experience value), the face image is discarded to improve the accuracy of identity recognition in the tracking process.

When a person is walking, the face detection module of the system may fail at any time due to the changeable posture. This will cause discontinuities in tracking of person. Therefore,

we add a human body detection module on the basis of face detection, which greatly increases the continuity of system's tracking and re-recognition.

The continuity of face detection and feature extraction is poor, but the accuracy of face recognition is extremely high; the continuity of human body detection and feature extraction is strong, but the personal identity information cannot be obtained. Therefore, the person tracking scheme designed in this paper uses face recognition to determine the identity of the person, and uses real-time person detection to track person.

Face detection and human body detection are independent modules. How to transfer the identity information obtained by face recognition to the human body so that it can track and determine the identity of the person through the human body is a problem that needs to be solved urgently. In response to this problem, we have designed Algorithm 1 to calculate the intersection ratio iou of all person face images and human body images, and get the matrix IOU. Then use the method proposed in *section C* to process the IOU matrix to obtain the matching pair of the face image and the human body image. In this way, the identity information obtained based on the face can be transferred to the body.

Algorithm 1 Personal identity information transfer

```
f_boxes is a list of face rectangle
p_boxes is a list of person rectangle
f_identities is a list of face identity information
p_identities is a list of body identity information
for all f_box, p_box in f_boxes, p_boxes do:
    get f_box's top left vertex and lower right vertex
    coordinates:  $(fx_1, fy_1), (fx_2, fy_2)$ 
    get p_box's top left vertex and lower right vertex coordinates:
     $(px_1, py_1), (px_2, py_2)$ 
     $ix_1 = \max(fx_1, px_1), ix_2 = \min(fx_2, px_2)$ 
     $iy_1 = \max(fy_1, py_1), iy_2 = \max(fy_2, py_2)$ 
     $iw = \max(0, (ix_2 - ix_1)), ih = \max(0, (iy_2 - iy_1))$ 
     $iarea = iw \times ih$ 
     $boxes2area = (px_2 - px_1) \times (py_2 - py_1)$ 
     $iou = 1 - iarea \div boxes2area$ 
end for
```

E. Person re-identification based on human body characteristics

We have designed Algorithm 2 to apply pedestrian re-recognition technology to the research of real-time person tracking and re-recognition under a single camera.

In the process of tracking person, we use the human body detection and shape feature extraction method in *section B* to maintain a real-time updated person information cache:

p_cache. In *p_cache*, each person will store his/her identity information, ID and 10 different moments of human body images. The identity information of the person body in *p_cache* is obtained by the method proposed in *section D* to realize the re-identification of person based on the characteristics of the human body. The Match function is implemented by the method proposed in *section C*.

Algorithm 2 Person re-identification

```
p_track is a set of (ID, t_images)
p_cache is a set of (p_ID, c_images, p_identity)
for all track in p_track do:
    add detected person images of current frame to p_track.t_images
    match_list  $\leftarrow$  Match(track.t_images, p_cache)
    for all match in match_list do:
        if match.distance < threshold:
            track.ID  $\leftarrow$  match.p_ID
        else:
            add new ID to p_cache.p_ID
            add match.t_image to p_cache.c_images
        end if
    end for
end for
```

III. SYSTEM IMPLEMENTATION AND EXPERIMENTAL EVALUATION

This section is divided into two parts. The first part introduces the *personTrack* system implemented by the person tracking and re-recognition scheme based on the matching of face and appearance features; The second part is the experimental part. This experiment compares the re-recognition accuracy and real-time performance of *personTrack*, paper[7] and *personTrackNoPF*(remove the re-identification module based on human body characteristics).

A. System introduction

Our system is developed with Python3 and PyQt5, and the video surveillance is recorded by Hikvision webcam in real time and transmitted to the server. Our system is deployed on a GPU server equipped with two NVIDIA TITAN Xp graphics cards. Our system(see Figure 2) can process the real-time video stream transmitted by the web camera, and can also process the offline surveillance video.

In the process of person tracking, situations such as face loss, person leaving the field of view, overlapping person bodies, and person ID exchanges often occur. This system can better deal with those emergencies, and realizes the function of accurately counting the online time of person in video and real-time video streams. For each frame of the detection, each person will be framed in a rectangular region, and the person's name and ID will be displayed above the rectangular region. A black area is spliced on the right side of the video frame to display the name, ID, and online duration.

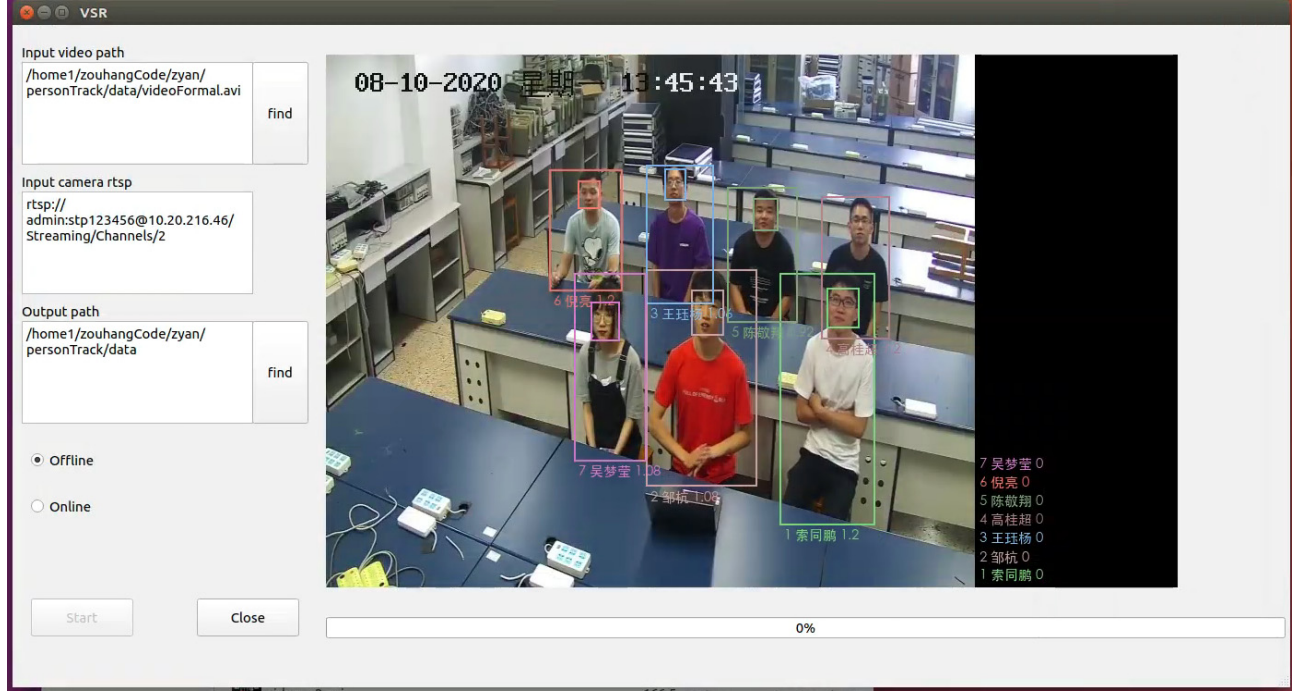


Figure 2. personTrack system.

B. Person re-identification experiment

In this paper, person re-identification experiments were carried out in different time periods. Use Haikang DS-2DC7423IW-A webcam to record 2 surveillance videos with lengths of 71 seconds and 253 seconds on different dates, named video1 and video2 respectively. There are 19 different person appeared in two surveillance videos. There are 7 person in video1, which are recorded as: M, N, O, P, Q, R, S; there are 13 person in video2, which are recorded as A, B, C, D, E, F, G, H, I, J, K, L, M. In video1 and video2, people often leave the monitoring area, and often appear emergencies such as missing faces and overlapping person. The number of face loss in video1 is 67 times, and the number of person loss is 779 times; the number of face loss in video2 is 6675 times, and the number of person loss is 562 times. Experiments with these surveillance videos can effectively verify the high continuity and accuracy of the *personTrack* system for person tracking and re-identification.

When the *personTrack* system runs on the GPU server, it takes about 50ms to obtain the human body detection rectangular per frame, and it takes about 10ms to obtain the face detection frame per frame. The appearance feature extraction takes about 15ms per frame, and the face feature extraction takes about 5ms per frame. In order to ensure real-time performance, when *personTrack* processes the surveillance video of person, it first extracts frames from the video, and then performs person re-identification and tracking tasks on the frames.

Table 1. RE-IDENTIFICATION ACCURACY

method	Person	Success	Unknown	accuracy
Tracking system	6	60	46	56.1%
<i>personTrack</i>	7	1911	204	90.4%
<i>personTrackNoPF</i>	7	714	69	91.2%
<i>personTrack</i>	13	15768	1313	83.6%
<i>personTrackNoPF</i>	13	7270	290	85.6%

This experiment extracts the information of each frame of surveillance video processed by *personTrack* and *personTrackNoPF* to calculate the re-identification accuracy and real-time tracking performance. Table 1 statistics the success times and unknown times of video1 and video2. $accuracy = success / re-recognition\ times$. Compared with the paper[7], when the number of people increases to 7, the re-identification accuracy rates of *personTrack* and *personTrackNoPF* increase from 56.1% of paper [7] to 90.4% and 91.2% respectively. When the number of person is increased to 13, the frequency of person occlusion and face loss is greatly increased, and the re-recognition accuracy rates of *personTrack* and *personTrackNoPF* can still reach 83.6% and 85.6%.

In terms of accuracy only, *personTrackNoPF* is 1 to 2% higher than *personTrack*. This is because, on the one hand, *personTrack* captures human body more than twice as many times as *personTrackNoPF* captures face, which causes a sharp increase in the number of re-identification times, resulting in an accuracy rate decreasing. On the other hand, the accuracy of face recognition is higher than body appearance recognition.

As shown in Table 2, real-time performance = $re-id\ times / video\ duration / person\ number$. The real-time tracking performance of paper [7] is only 0.047, and the real-time tracking performance of *personTrack* and *personTrackNoPF* are 4.997 and 2.079 respectively, which are 106 times and 44 times higher than paper [7]. *PersonTrack*'s real-time performance is 2.4 times higher than *personTrackNoPF* without reducing the accuracy of re-recognition.

Table 2. REAL-TIME TRACKING PERFORMANCE

Method	person number	re-id times	video duration	performance
Tracking system[7]	6	107	380	0.047
personTrack	7	2115	71	4.256
personTrackNoPF	7	783	71	1.575
personTrack	13	18868	253	5.737
personTrackNoPF	13	8492	253	2.582

IV. CONCLUSION

This paper has proposed a multi-person real-time tracking and re-recognition scheme based on human body feature matching. Based on high-accuracy face recognition, combined with reliable person appearance recognition, it can recognize person with high accuracy. At the same time, maintain high continuity of person identity tracking. During tracking, a real-time updated cache list of person information is established. Even if a person leaves the camera for a period of time, when he enters the monitoring view again, he can still be re-identified. The experimental part of this paper evaluated the re-recognition accuracy and real-time tracking performance of the *personTrack* system. Compared with the paper [7], the accuracy and real-time tracking performance of *personTrack* still have a significant improvement when the number of tracking person increases.

Tracking and re-identification for a large number of person is still a difficult problem. The caching mechanism proposed in this paper will reduce the efficiency of system performance when facing a large number of people. In addition, as the

number of people in surveillance videos increases, the re-identification accuracy rate will gradually decrease.

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